12A 1 INTRODUCTION TO ENTROPY

SPECIFICATION REFERENCE
12.1 12.2
12.3 12.4

LEARNING OBJECTIVES

- Understand that, since endothermic reactions can occur spontaneously at room temperature, enthalpy changes alone do not control whether reactions occur.
- Understand entropy as a measure of disorder of a system in terms of the random dispersal of molecules and of energy quanta between molecules.
- Understand that the entropy of a substance increases with temperature, that entropy increases as solid → liquid → gas and that perfect crystals at zero kelvin have zero entropy.
- Be able to interpret the natural direction of change as being in the direction of increasing total entropy (positive entropy change), including gases spread spontaneously through a room.

WHAT MAKES A REACTION OCCUR?

Perhaps one of the most important questions to ask in chemistry is 'Will a reaction occur?'

We know that, once started, some chemical reactions simply 'go' with no further, continuous help from us.

For example, ammonia gas and hydrogen chloride gas react together at room temperature to form the white solid, ammonium chloride:

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Magnesium, once ignited, will burn in oxygen to form magnesium oxide:

$$Mg(s) + \tfrac{1}{2}O_2(g) \rightarrow MgO(s)$$

In other reactions, rather than the reactants changing completely into the products, a position of equilibrium is reached, with the final mixture containing a measurable amount of both reactants and products.

For example, ethanoic acid dissociates in water. In a 0.1 mol dm^{-3} solution of ethanoic acid, only about 1% of the ethanoic acid molecules are present as ions:

$$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$$

Another example is the dimerisation of nitrogen dioxide in the gas phase:

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

At a temperature of 298 K and a pressure of 100 kPa, the equilibrium mixture contains about 70% N_2O_4.

There are other types of reactions that simply do not occur at all, at least not without some help. For example:

- ammonium chloride does not spontaneously decompose into ammonia and hydrogen chloride
- magnesium oxide does not break apart to form magnesium and oxygen without some continuous intervention from us in the form of heating.

If we consider all of these types of reaction together, we see that the real difference between them is the *position* of equilibrium that is established. For some reactions, the position of equilibrium is so far over to the products side that, to all intents and purposes, the reaction has gone to completion. For some other reactions, significant amounts of both reactants and products are present at equilibrium. For other reactions, the equilibrium lies so far to the left that they appear not to take place at all.

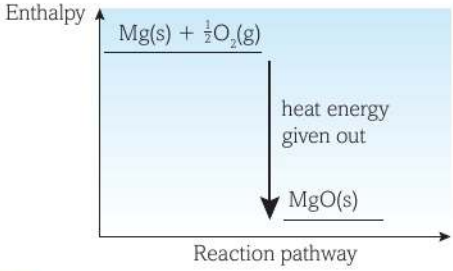
So, perhaps the better question to ask is *not* 'Will a chemical reaction occur?' but 'What will be the position of equilibrium?' This is the question we hope to answer in this topic.

DID YOU KNOW?

A campfire is an example of increases in entropy. The solid wood burns and becomes ash, smoke and gases, all of which spread energy outwards more easily than the solid fuel.

EXOTHERMIC AND ENDOTHERMIC REACTIONS

The reaction between magnesium and oxygen is exothermic and can be represented by the enthalpy level diagram shown in **fig A**.



▲ **fig A** Enthalpy level diagram for the reaction between magnesium and oxygen.

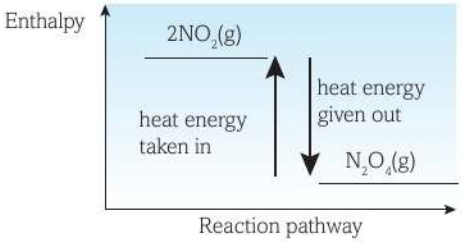
As the products have less energy than the reactants, we often say that the products are more *energetically stable* than the reactants.

It is very tempting to conclude that this reaction occurs *because* the magnesium oxide is energetically more stable than its elements, magnesium and oxygen. However tempting this argument may be, we must discard it, because experience tells us that many endothermic reactions occur at room temperature.

Let us consider the dimerisation of $NO_2(g)$ at 298 K:

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

The enthalpy level diagram for this reaction is shown in **fig B**.



▲ **fig B** Enthalpy level diagram for the dimerisation of NO_2.

This reaction is readily reversible and reaches a position of equilibrium at 298 K. The forward reaction leading to the formation of $N_2O_4(g)$ is exothermic. However, the backward reaction is endothermic.

As we discovered in **Topic 9** (**Book 1: IAS**), the position of equilibrium can be reached from either direction. This means that if we place some $N_2O_4(g)$ in a sealed container at room temperature, some of it will decompose to form $NO_2(g)$. So, an endothermic reaction is taking place at room temperature without us doing anything to it. We say that the reaction is *spontaneous* (see later in this section).

Clearly, it is not only exothermic reactions that can take place spontaneously. The driving force for spontaneous endothermic reactions cannot be the formation of more energetically favourable, i.e. lower-energy, products but must involve another factor we have yet to consider.

That factor is a quantity known as **entropy**, which is governed by the *Second Law of Thermodynamics*.

But first, we need to appreciate what is meant by the term **spontaneous process**.

SPONTANEOUS PROCESSES

A spontaneous process is one that takes place without continuous intervention by us.

A good example of a spontaneous process is the freezing of water to form ice. If water is placed in an environment at −20 °C it will turn into ice. However, the reverse will never happen: at −20 °C ice will never melt to form water.

Another example of a spontaneous process is the mixing of gases. If two unreactive gases are present in a container, they will mix completely in a process known as diffusion.

We can bring about the reverse of spontaneous processes by intervening. For example, we can melt ice by heating it, and we can separate a mixture of gases by liquefying followed by distillation.

The key point, however, is that the reverse of a spontaneous process never happens on its own. A mixture of gases will *never* separate on their own.

At the beginning of this section we stated that, in chemistry, we are concerned with whether or not chemical reactions will occur, and we suggested that perhaps the most important question to ask in chemistry is 'Why does a reaction occur?'

From the point of view of thermodynamics, all reactions are reversible, so we suggested that an even better question to ask is:

> When a reaction reaches a point of equilibrium, what determines whether the equilibrium will favour the reactants or the products, and to what extent?

This question is answered using the Second Law of Thermodynamics, in which the concept of entropy is introduced.

ENTROPY

The Second Law of Thermodynamics allows us to predict whether a process is likely to occur. It is, therefore, the key to understanding what drives chemical reactions and what determines the position of equilibrium.

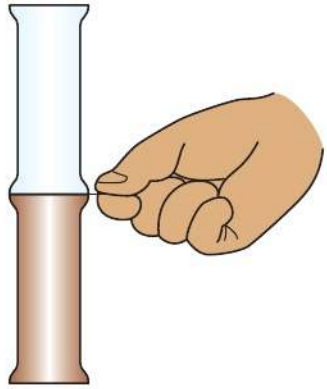
The Second Law of Thermodynamics introduces a term called entropy, which is a property of matter, just like density or energy. The simplest way to describe entropy is to say that it is a quantity associated with randomness or disorder.

This definition is rather basic and has to be used with care. Entropy refers not only to the distribution of molecules but also to the ways of distributing the energy of the system in all of the energy levels available.

To help you to understand entropy, it helps to think about what happens during the diffusion of gases.

DISTRIBUTION OF MOLECULES

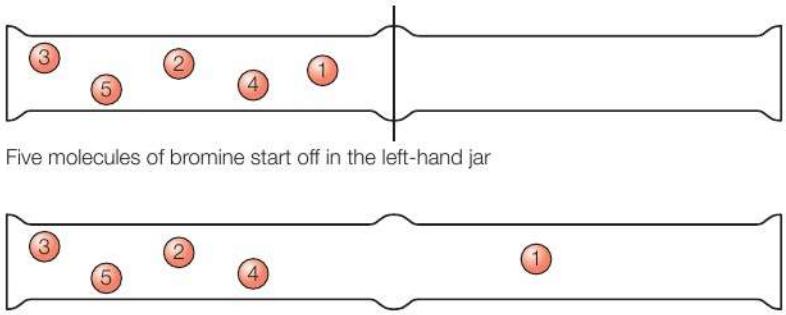
Fig C shows two gas jars. The bottom one contains bromine gas and the top one contains air.



▲ **fig C** Gas jars containing air and bromine

When the cover slip between the two gas jars is removed, the bromine and air diffuse so that the molecules of each gas are spread out evenly throughout both gas jars.

To understand why this happens we can look at a much simpler example where there are only five molecules. **Fig D** shows two gas jars. The left-hand jar contains five molecules of bromine and the right-hand jar contains no molecules.



Five molecules of bromine start off in the left-hand jar

As soon as the cover slip is removed, the molecules are free to move between the jars

▲ **fig D** Diffusion in a gas: one of the ways the molecules might rearrange. Remember that air molecules are also present but have not been shown.

Each molecule has two possible arrangements. It can be in either the left-hand jar or the right-hand jar. We can write the number of possible arrangements for molecule 1 as W_1, and the number of possible arrangements for molecule 2 as W_2, and so on.

As there are two possible arrangements for molecule 1, $W_1 = 2$. For molecule 2, $W_2 = 2$, and so on.

The equation for the *total* number of ways, W, that the five molecules can be arranged is given by:

$$\begin{aligned} W &= W_1 \times W_2 \times W_3 \times W_4 \times W_5 \\ &= 2 \times 2 \times 2 \times 2 \times 2 \\ &= 2^5 \\ &= 32 \end{aligned}$$

Most of the 32 possible arrangements have molecules in both jars. Only one has all five molecules in the left-hand jar. It would be surprising if this arrangement happened very often.

If we increase the number of molecules to 100, W becomes 2^{100}. The left-hand gas jar would contain all 100 molecules only once in every 3.2×10^{22} years! We would have to wait for a very long time indeed to see all 100 molecules in the left-hand jar!

You can see that as the number of molecules increases, the number of possible arrangements also increases very rapidly. To make the numbers easier to work with we need to use an equation to help us. You do not need to remember the equation, but it helps us to see how the spreading out of the molecules is related to an increase in entropy.

Entropy, S, is linked to W by the equation

$$S = k \ln W$$

where W is the number of ways of arranging the molecules, k is a constant called the Boltzmann constant, with a value $1.38 \times 10^{-23}\,\text{J K}^{-1}$ and ln is the natural logarithm (i.e. the logarithm to the base e).

Entropy is measured in units of $\text{J K}^{-1}\,\text{mol}^{-1}$.

Entropy measures the degree of randomness, so you would expect a gas to have a larger entropy than its liquid form, and a liquid to have a larger entropy than its solid form. The figures in **table A** show that this is the case for water vapour, water and ice.

SUBSTANCE	ENTROPY, $S/\text{J K}^{-1}\,\text{mol}^{-1}$
water vapour	189
water	70
ice	48

table A Entropies of water vapour, water and ice.

EXAM HINT

It is satisfying to make links between topics in chemistry. If you multiply the Boltzmann constant k (1.38 × 10–23) by the Avogadro constant L (6.02 × 1023),
you will get 8.31. This is also known as the gas constant, R, which you should remember from PV = nRT in **Topic 1 (Book 1: IAS)**.

DISTRIBUTION OF ENERGY

The spreading out of molecules in diffusion is an increase in entropy, because the molecules become more randomly dispersed. In the same way, the spreading out of heat energy also represents an increase in entropy.

Energy exists in 'packets' called quanta. You can have a whole number of quanta, but not halves or fractions of quanta.

The distribution of the quanta between molecules is random. The more quanta there are, the more ways there are of arranging them between the molecules. Also, the more molecules there are, the more ways there are of sharing the quanta.

As an example, **table B** shows that there are five possible arrangements of four quanta between two molecules. **Table C** shows that there are six possible arrangements of five quanta between two molecules. **Table D** shows that there are 10 possible arrangements of three quanta between three molecules.

table B

MOLECULE 1	MOLECULE 2
4	0
3	1
2	2
1	3
0	4

table C

MOLECULE 1	MOLECULE 2
5	0
4	1
3	2
2	3
1	4
0	5

table D

MOLECULE 1	MOLECULE 2	MOLECULE 3
3	0	0
0	3	0
0	0	3
2	1	0
2	0	1
1	1	1
1	2	0
1	0	2
0	1	2
0	2	1

table B 2 molecules, 4 quanta, 5 arrangements
table C 2 molecules, 5 quanta, 6 arrangements
table D 3 molecules, 3 quanta, 10 arrangements

The higher the temperature, the more energy a substance has. The more energy a substance has, the more ways there are to distribute the energy. The more ways there are to distribute the energy, the higher the entropy. So, increasing the temperature will increase the entropy.

PERFECT CRYSTALS

A 'perfect' crystal is one in which the internal lattice structure is the same at all times. In other words, the particles (atoms, molecules or ions) are fixed and are not moving in any way, rotating or vibrating.

The Third Law of Thermodynamics states that the entropy of a perfect crystal at the temperature of absolute zero (i.e. 0 K) is zero.

THE SECOND LAW OF THERMODYNAMICS

The Second Law of Thermodynamics is sometimes misquoted. You may have come across the statement that in a spontaneous change, entropy always increases. This cannot be true: if it were, water could never freeze because this change involves a decrease in entropy.

Another example is the reaction between ammonia gas and hydrogen chloride gas:

$$NH_3(g) + HCl(g) \rightarrow NH_4Cl(s)$$

Clearly, there is a reduction in the entropy of the system, since two gases are forming a solid, yet this reaction occurs spontaneously at room temperature.

The statement *entropy always increases* is almost correct. It needs to be expressed more carefully to give the Second Law of Thermodynamics:

In a spontaneous process, the *total* entropy increases.

In **Section 12A.2**, we explain what is meant by the term 'total entropy'.

CHECKPOINT

1. For each of the following, predict whether the change is accompanied by an increase or a decrease in total entropy.
   (a) $H_2O(g) \rightarrow H_2O(l)$ (b) $I_2(s) \rightarrow I_2(g)$ (c) $Na(l) \rightarrow Na(s)$
2. Arrange the following in order of entropy, with the lowest first:
   steam at 110 °C; ice at −10 °C; water at 30 °C

SUBJECT VOCABULARY

entropy a property of matter that is associated with the degree of disorder, or degree of randomness, of particles (i.e. atoms, molecules or ions), and also with the distribution of the quanta of energy between the particles

spontaneous process a process that takes place without continuous intervention by us

12A 2 TOTAL ENTROPY

SPECIFICATION REFERENCE

12.6 | 12.7 | 12.8 | 12.9(i) | 12.9(ii) | 12.10 | 12.11

LEARNING OBJECTIVES

- Understand that the total entropy change of any reaction is the sum of the entropy change of the system and the entropy change of the surroundings, summarised by the expression:
  $\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$
- Be able to calculate the entropy change of the system for a reaction, ΔS_{system}, given the entropies of the reactants and products.
- Be able to calculate the entropy change of the surroundings, and hence ΔS_{total}, using the expression
  $\Delta S_{surroundings} = \frac{\Delta H}{T}$
- Understand that the feasibility of a reaction depends on:
  (i) the balance between ΔS_{system} and $\Delta S_{surroundings}$, so that even endothermic reactions can occur spontaneously at room temperature
  (ii) temperature, as higher temperatures decrease the magnitude of $\Delta S_{surroundings}$, so its contribution to ΔS_{total} is less.
- Understand that reactions can occur as long as ΔS_{total} is positive even if one of the other entropy changes is negative.
- Understand and distinguish between the concepts of thermodynamic stability and kinetic stability.

TOTAL ENTROPY CHANGE, ΔS_{total}

The total entropy change of a process comprises two components:

- the entropy change of the system, ΔS_{system}, and
- the entropy change of the surroundings, $\Delta S_{surroundings}$.

In a chemical reaction, the system is the species that are taking part in the reaction.

The surroundings is everything else. In practice, this usually means the reaction vessel, e.g. test tube or beaker, and the air in the laboratory.

The total entropy change is defined as the sum of the entropy change of the system and the entropy change of the surroundings. That is:

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$$

For a reaction to be spontaneous, ΔS_{total} must be positive. This is another way of expressing the Second Law of Thermodynamics.

DID YOU KNOW?

The official definition of the Second Law of Thermodynamics does not use the term 'total entropy change'. Instead, it refers to the 'entropy change of the universe'. These two terms have the same meaning. We will use the term 'total entropy change' throughout this book.

ENTROPY CHANGE OF THE SYSTEM

ΔS_{system} is calculated using the expression:

$$\Delta S_{system} = \Sigma S\,(\text{products}) - \Sigma S\,(\text{reactants})$$

where S represents entropy and Σ represents 'the sum of'.

The standard entropy values of some substances are shown in **table A**.

Standard refers to conditions of 100 kPa and 298 K.

GAS	ENTROPY $S^\ominus$ /J K^{-1} mol^{-1}
$H_2O(g)$	188.7
H_2	130.6
O_2	205.0
N_2	191.6
Cl_2	165.0
CO_2	213.6
NH_3	192.3

LIQUID	ENTROPY $S^\ominus$ /J K^{-1} mol^{-1}
$H_2O(l)$	69.9
CH_3OH	239.7
CH_3CH_2OH	160.7
C_6H_6	172.8

SOLID	ENTROPY $S^\ominus$ /J K^{-1} mol^{-1}
$H_2O(s)$	47.9
C(diamond)	2.4
C(graphite)	5.7
CaO	39.7
$CaCO_3$	92.9

table A

EXAM HINT

An examiner may ask you to **suggest** why the standard entropy value of diamond is lower than that of graphite. Diamond is ordered in three dimensions (a 3D lattice), whereas graphite is ordered in two dimensions and has greater degrees of freedom between the layers.

WORKED EXAMPLE 1

Use the values in **table A** to calculate $\Delta S^\ominus_{system}$ for the following reaction:

$$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$$

$\Delta S^\ominus_{system} = S^\ominus[CaO(s)] + S^\ominus[CO_2(g)] - S^\ominus[CaCO_3(s)]$

$= 39.7 + 213.6 - 92.9$

$= +160.4\ J\ K^{-1}\ mol^{-1}$

WORKED EXAMPLE 2

Use the values in **table A** to calculate $\Delta S^\ominus_{system}$ for the following reaction:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

$\Delta S^\ominus_{system} = 2 \times S^\ominus[NH_3(g)] - S^\ominus[N_2(g)] - 3 \times S^\ominus[H_2(s)]$

$= (2 \times 192.3) - 191.6 - (3 \times 130.6)$

$= -198.8\ J\ K^{-1}\ mol^{-1}$

ENTROPY CHANGE OF THE SURROUNDINGS

The entropy change of the surroundings, $\Delta S_{surroundings}$, is related to the enthalpy change of the reaction, ΔH, by the expression:

$$\Delta S_{surroundings} = -\frac{\Delta H}{T}$$

where T is the temperature in kelvin.

For an *exothermic* reaction, where ΔH is negative, $\Delta S_{surroundings}$ will always be positive, so the entropy of the surroundings *increases*.

Conversely, for an *endothermic* reaction, $\Delta S_{surroundings}$ will always be negative, so the entropy of the surroundings *decreases*.

WORKED EXAMPLE 3

Calculate $\Delta S_{surroundings}$ at 298 K when one mole of hydrogen gas is burned in oxygen.

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \qquad \Delta H = -286\ kJ\ mol^{-1}$$

$$\Delta S_{surroundings} = -\frac{-286}{298}$$

$= +0.960\ kJ\ K^{-1}\ mol^{-1}$ or $+960\ J\ K^{-1}\ mol^{-1}$

LEARNING TIP

Note that entropy values are usually quoted in J K^{-1} mol^{-1}, whereas enthalpy changes are usually quoted in kJ mol^{-1}. So, in any equation linking ΔS and ΔH, one of the energy terms will need a conversion of units.

CALCULATING THE TOTAL ENTROPY CHANGE, ΔS_{total}

Now that we know how to calculate both ΔS_{system} and $\Delta S_{surroundings}$, we are in a position to calculate the total entropy change, ΔS_{total}, for a reaction.

WORKED EXAMPLE 4

Using the information in **table A**, calculate the total entropy change at 298 K for the following reaction:

$$H_2(g) + \tfrac{1}{2}O_2(g) \rightarrow H_2O(l) \quad \Delta H = -286\ \text{kJ mol}^{-1}$$

$$\Delta S_{system} = S[H_2O(l)] - S[H_2(g)] - \tfrac{1}{2} \times S[O_2(g)]$$
$$= 69.9 - 130.6 - (\tfrac{1}{2} \times 205)$$
$$= -163.2\ \text{J K}^{-1}\ \text{mol}^{-1}$$
$$\Delta S_{total} = -163.2\ \text{J K}^{-1}\ \text{mol}^{-1} + 960\ \text{J K}^{-1}\ \text{mol}^{-1}$$
$$= +796.8\ \text{J K}^{-1}\ \text{mol}^{-1}$$

SUMMARY

ΔS_{total} will be positive if:

- both $\Delta S_{surroundings}$ and ΔS_{system} are positive
- $\Delta S_{surroundings}$ is positive and ΔS_{system} is negative, but the magnitude of $\Delta S_{surroundings}$ > the magnitude of ΔS_{system}
- $\Delta S_{surroundings}$ is negative and ΔS_{system} is positive, but the magnitude of $\Delta S_{surroundings}$ < the magnitude of ΔS_{system}.

THERMODYNAMIC AND KINETIC STABILITY

The reaction between methane and oxygen is exothermic, and the total entropy change has a positive value:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \quad \Delta H_r^{\ominus} = -890.3\ \text{kJ mol}^{-1}$$
$$\Delta S_{total} = +2.95\ \text{kJ mol}^{-1}$$

Since ΔS_{total} is positive, the reaction is thermodynamically feasible at 298 K. However, the reaction does not take place until the reaction mixture is ignited, usually with a flame.

The added heat energy is required to overcome the high activation energy of the reaction.

A reaction mixture with a high activation energy, where a reaction will not take place spontaneously at room temperature, is said to be kinetically stable.

You can see here that a reaction mixture (such as methane and oxygen) can be thermodynamically unstable but kinetically stable, so a reaction does not take place under standard conditions.

THE ROLE OF TEMPERATURE

The increase in entropy obtained by supplying a certain amount of heat energy to an object depends on the temperature of the system.

If an object is very cold, the molecules are not moving around very much. Supplying some heat energy to the object will make the molecules move around more, so the entropy increases.

If we supply the same amount of heat energy to a much hotter object, the entropy will still increase, but not by as much as with the cold object. This is because in the hot object the molecules are already moving around vigorously and the increased degree of movement is less for the hot object.



▲ **fig A** Why does water freeze?

WHY DOES WATER FREEZE?

We will now apply these principles to explain why, under certain conditions, water will freeze.

Ice has lower entropy than liquid water, so ΔS_{system} is *negative*.

The process is exothermic, so $\Delta S_{surroundings}$ is *positive*.

If the magnitude of $\Delta S_{surroundings}$ > the magnitude of ΔS_{system}, then ΔS_{total} is *positive* and the water will freeze.

We will now calculate ΔS_{total} for the change of water into ice at +5 °C and −5 °C using the following data.

$H_2O(l) \rightarrow H_2O(s) \qquad \Delta H = -6010\ J\ mol^{-1}$

- $S^{\ominus}$(water) = 69.9 J K^{-1} mol^{-1}
- $S^{\ominus}$(ice) = 47.9 J K^{-1} mol^{-1}

At +5 °C (278 K):

$$\Delta S_{system} = (47.9 - 69.9) = -22.0\ J\ K^{-1}\ mol^{-1}$$

$$\Delta S_{surroundings} = -\frac{-6010}{278} = +21.6\ J\ K^{-1}\ mol^{-1}$$

$$\Delta S_{total} = -0.4\ J\ K^{-1}\ mol^{-1}$$

The total entropy change is negative, meaning that the change is not thermodynamically spontaneous. The water will not freeze.

At −5 °C (268 K):

$$\Delta S_{system} = (47.9 - 69.9) = -22.0\ J\ K^{-1}\ mol^{-1}$$

$$\Delta S_{surroundings} = -\frac{-6010}{268} = +22.4\ J\ K^{-1}\ mol^{-1}$$

$$\Delta S_{total} = +0.4\ J\ K^{-1}\ mol^{-1}$$

The total entropy change is positive, meaning that the change is thermodynamically spontaneous. The water will freeze.

It is interesting to note that what has changed between +5 °C and −5 °C is the entropy of the surroundings. The reason that $\Delta S_{surroundings}$ has changed in magnitude is simply because of the temperature change. The same amount of heat energy has been transferred to the surroundings but, because the entropy of the surroundings has a higher temperature when at +5 °C than when at −5 °C, the change in entropy is smaller, as explained earlier.

So, the reason we put water into a freezer when we want to make ice is because the entropy change of the surroundings (i.e. essentially the air inside the freezer) is large enough to compensate for the decrease in entropy when the water freezes.

CHECKPOINT

SKILLS CREATIVITY

1. Hydrated cobalt(II) chloride dehydrates on heating according to the following equation:

$CoCl_2.6H_2O(s) \rightarrow CoCl_2(s) + 6H_2O(l) \quad \Delta H^{\ominus}_{298\ K} = +88.1\ kJ\ mol^{-1}$

(a) Calculate the standard entropy change of the system, $\Delta S^{\ominus}_{system}$.

$S^{\ominus}[CoCl_2.6H_2O(s)] = 343.0\ J\ K^{-1}\ mol^{-1}$

$S^{\ominus}[CoCl_2(s)] = 109.2\ J\ K^{-1}\ mol^{-1}$

$S^{\ominus}[H_2O(l)] = 69.9\ J\ K^{-1}\ mol^{-1}$

(b) Calculate the standard entropy change of the surroundings, $\Delta S^{\ominus}_{surroundings}$, at 298 K.

(c) Calculate the standard total entropy change, $\Delta S^{\ominus}_{total}$, at 298 K for this reaction.

(d) Explain whether hydrated cobalt(II) chloride can be stored at 298 K without it dehydrating?

2. Use the data provided to calculate the total standard entropy change, $\Delta S^{\ominus}_{total}$, at 298 K, for the following reaction:

$2Fe(s) + 1\frac{1}{2}O_2(g) \rightarrow Fe_2O_3(s)$

$\Delta H^{\ominus} = -822\ kJ\ mol^{-1}$

$S^{\ominus}[Fe(s)] = 27.2\ J\ K^{-1}\ mol^{-1}$

$S^{\ominus}[O_2(g)] = 205.0\ J\ K^{-1}\ mol^{-1}$

$S^{\ominus}[Fe_2O_3(s)] = 90.0\ J\ K^{-1}\ mol^{-1}$

12A 3 UNDERSTANDING ENTROPY CHANGES

SPECIFICATION REFERENCE
12.5(i) 12.5(ii) 12.5(iii) 12.19

LEARNING OBJECTIVES

- Understand why entropy changes occur during:
  (i) changes of state
  (ii) dissolving of a solid ionic lattice
  (iii) reactions in which there is a change in the number of moles from reactants to products
- Be able to use entropy and enthalpy changes of solution values to predict the solubility of ionic compounds and discuss trends in the solubility of ionic compounds covered in Unit 2

REACTIONS INVOLVING A CHANGE OF STATE

We have already mentioned that, in general, entropy increases in the order:

solid < liquid < gas

So, we would expect an increase in the entropy of the system if a gas is produced from a reaction involving a solid and/or a liquid.

EXAMPLE 1

When solid ammonium carbonate is added to pure ethanoic acid, bubbles of gas are rapidly produced. Despite its violent appearance, this is an endothermic reaction, as can be shown by placing a thermometer in the acid before the ammonium carbonate is added. The temperature falls considerably as the reaction takes place.

$$2CH_3COOH(l) + (NH_4)_2CO_3(s) \rightarrow 2CH_3COONH_4(aq) + H_2O(l) + CO_2(g)$$

Since the reaction is endothermic, $\Delta S_{surroundings}$ will be negative. However, there is a large increase in the entropy of the system, ΔS_{system}, because a gas is produced from a liquid and a solid. The magnitude of ΔS_{system} is greater than that of $\Delta S_{surroundings}$, and this makes ΔS_{total} positive, so the reaction is thermodynamically spontaneous.

EXAMPLE 2

Hydrated barium hydroxide reacts with solid ammonium chloride in a rapid endothermic reaction at room temperature:

$$Ba(OH)_2.8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2(s) + 10H_2O(l) + 2NH_3(g)$$

As in the previous example, the driving force of the reaction is ΔS_{system}, which overcomes the negative value of $\Delta S_{surroundings}$ caused by the endothermic nature of the reaction. The reactants are solids and the products are a solid, a liquid and a gas.

EXAMPLE 3

Magnesium burns in oxygen to form solid magnesium oxide:

$$Mg(s) + \tfrac{1}{2}O_2(g) \rightarrow MgO(s)$$

The reaction is highly exothermic, so $\Delta S_{surroundings}$ will be positive. ΔS_{system} is negative since a solid and a gas are changing into a solid. However, the magnitude of the value of $\Delta S_{surroundings}$ is greater than that of ΔS_{system}, making ΔS_{total} positive. So, the reaction is thermodynamically spontaneous.

DISSOLVING IONIC SOLIDS IN WATER

When an ionic solid dissolves in water, two changes take place:

- the lattice structure is broken down, and
- the ions become hydrated.

The breaking down of the lattice structure is an endothermic process, equivalent to the reverse of the lattice energy. It also results in an increased number of moles of particles present, so increases the entropy of the salt.

The hydration of the ions is an exothermic process, but results in the water molecules becoming more ordered as they arrange themselves in an orderly manner around the positive and negative ions. The increase in ordering of the water molecules produces a decrease in entropy of the water. This ordering of the water molecules is particularly significant when dissolving anhydrous solids in water.

To explain why some ionic solids are soluble in water, while others are insoluble, we need to look at both the enthalpy and the entropy changes involved.

The solubility of an ionic solid is determined by the total entropy change for the solid.

$$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings}$$

Since $\Delta S_{surroundings} = -\dfrac{\Delta_{sol}H}{T}$

this expression becomes:

$$\Delta S_{total} = \Delta S_{system} - \frac{\Delta_{sol}H}{T}$$

The value of ΔS_{total}, and hence the solubility of the solid, depends on the values of three factors:

- the entropy change of the system, ΔS_{system}
- the enthalpy change of solution, $\Delta_{sol}H$
- the temperature, in kelvin, of the water, T.

Let us have a close look at the dissolving of ammonium nitrate crystals in water at a temperature of 298 K.

$NH_4NO_3(s) \xrightarrow{aq} NH_4^+(aq) + NO_3^-(aq)$

$\Delta_{sol}H = +25.8\,kJ\,mol^{-1}$

$\Delta S_{system} = S[NH_4^+(aq)] + S[NO_3^-(aq)] - S[NH_4NO_3(s)]$

$= +113.4 + 146.4 - 151.1\,J\,K^{-1}\,mol^{-1}$

$= +108.7\,J\,K^{-1}\,mol^{-1}$

$\Delta S_{surroundings} = -\frac{\Delta H_{sol}}{T} = -\frac{+25800}{298} = -86.6\,J\,K^{-1}\,mol^{-1}$

$\Delta S_{total} = (+108.7 - 86.6) = +22.1\,J\,K^{-1}\,mol^{-1}$

Since ΔS_{total} is positive, the dissolving of ammonium nitrate in water at 298 K is thermodynamically spontaneous. Since the activation energy for this process is very low, ammonium chloride is soluble in water at 298 K.

We are now in a position to think about the solubility in water of some other ionic solids.

Table A shows the relevant thermodynamic data for some solids at a temperature of 298 K. The values have been quoted to the nearest whole number.

IONIC SOLID	$\Delta_{sol}H$/kJ mol^{-1}	$\Delta S_{surroundings}$/J K^{-1} mol^{-1}	ΔS_{system}/J K^{-1} mol^{-1}	ΔS_{total}/J K^{-1} mol^{-1}	SOLUBILITY
NaCl	+4	−13	+43	+56	soluble
NH_4Cl	+15	−50	+167	+117	soluble
AgCl	+66	−221	+33	−188	insoluble
$MgSO_4$	−91	+305	−213	+92	soluble
$CuSO_4$	−73	+245	−192	+53	soluble
$CaSO_4$	−18	+60	−145	−85	insoluble

table A

REACTIONS INVOLVING A CHANGE IN NUMBER OF MOLES FROM REACTANTS TO PRODUCTS

If you increase the number of moles that you have, you automatically increase the number of particles (i.e. atoms, molecules or ions) present. This will result in an increase in the number of ways that the particles can be arranged, and this increases the entropy of the system, making ΔS_{system} positive.

We will now look again at the reactions shown above, but this time we will consider the change in the number of moles from reactants to products.

EXAMPLE 1

$2CH_3COOH(l) + (NH_4)_2CO_3(s) \rightarrow 2CH_3COONH_4(s) + H_2O(l) + CO_2(g)$

Number of moles of reactants = 3

Number of moles of products = 4

So, ΔS_{system} is positive.

EXAMPLE 2

$Ba(OH)_2.8H_2O(s) + 2NH_4Cl(s) \rightarrow BaCl_2(s) + 10H_2O(l) + 2NH_3(g)$

Number of moles of reactants = 3

Number of moles of products = 13

So, ΔS_{system} is positive.

DID YOU KNOW?

When we state that the solids AgCl and $CaSO_4$ are insoluble in water, this is a simplification. Both solids have a measurable solubility in water, but the solubility is very small. What we are really saying is that the position of the equilibrium in both cases is very far over to the solid side of the equation. We will come back to this point in **Topic 13**, when we discuss the relationship between the total entropy change for a reaction and the equilibrium constant, *K*, for the reaction.

EXAMPLE 3

$$Mg(s) + \tfrac{1}{2}O_2 \rightarrow MgO(s)$$

Number of moles of reactants = 1.5

Number of moles of products = 1

So, ΔS_{system} is negative.

SOLUBILITY TRENDS IN GROUP 2

In **Topic 8 (Book 1: IAS)**, we mentioned that for Group 2 the solubility of the metal hydroxides increases down the group and the solubility of the metal sulfates decreases down the group. We can now explain the reason for these trends in solubility.

As we have mentioned earlier, the two factors that affect the solubility of ionic compounds are the standard enthalpy change of solution of the compound, $\Delta_{sol}H^{\ominus}$, and the standard entropy change of the system, $\Delta S^{\ominus}_{system}$.

It is difficult to work out $\Delta S^{\ominus}_{system}$ accurately for many ionic compounds. We can use the standard entropy values of the ions as a guide.

SOLUBILITY OF GROUP 2 METAL HYDROXIDES

Table B shows the standard enthalpy and standard entropy changes involved in dissolving the Group 2 metal hydroxides. The values are given here to the nearest whole number.

METAL HYDROXIDE	$\Delta H^{\ominus}_{sol}$	$\Delta S^{\ominus}_{surroundings}$ / $JK^{-1}\,mol^{-1}$	HYDRATED CATION	STANDARD ENTROPY OF HYDRATED CATION / $JK^{-1}\,mol^{-1}$
$Mg(OH)_2$	+3	−10	$Mg^{2+}(aq)$	−138
$Ca(OH)_2$	−16	+54	$Ca^{2+}(aq)$	−53
$Sr(OH)_2$	−46	+154	$Sr^{2+}(aq)$	−33
$Ba(OH)_2$	−52	+174	$Ba^{2+}(aq)$	+10

table B

The values for the standard entropy of the hydroxide ion have been left out here, because the ion is common to all of the metal hydroxides. The trend in the standard entropies of the cations mirrors the trend in the standard entropy change of the system, $\Delta S^{\ominus}_{system}$.

The standard enthalpy change of solution becomes more negative down the group. This favours solubility, as $\Delta S^{\ominus}_{surroundings}$ becomes more positive.

The values of the entropies of the hydrated cations become less negative (i.e. more positive), and this also favours solubility.

As both factors favour an increase in solubility, the metal hydroxides become more soluble as you go down the group.

SOLUBILITY OF GROUP 2 METAL SULFATES

Table C shows the standard enthalpy and standard entropy changes involved in dissolving the Group 2 metal sulfates. The values are given here to the nearest whole number.

METAL SULFATE	$\Delta H^{\ominus}_{sol}$	$\Delta S^{\ominus}_{surroundings}$ / $JK^{-1}\,mol^{-1}$	HYDRATED CATION	STANDARD ENTROPY OF HYDRATED CATION / $JK^{-1}\,mol^{-1}$
$MgSO_4$	−91	+305	$Mg^{2+}(aq)$	−138
$CaSO_4$	−18	+60	$Ca^{2+}(aq)$	−53
$SrSO_4$	−9	+30	$Sr^{2+}(aq)$	−33
$BaSO_4$	+19	−63	$Ba^{2+}(aq)$	+10

table C

Again, the values for the standard entropy of the sulfate ion have been left out here, because the ion is common to all of the metal sulfates. The trend in the standard entropies of the cations mirrors the trend in the standard entropy change of the system, $\Delta S^{\ominus}_{system}$.

The standard enthalpy change of solution becomes less negative down the group. This favours insolubility, as $\Delta S^{\ominus}_{surroundings}$ becomes less positive.

The values of the entropy of the hydrated cation become less negative (i.e. more positive), and this also favours solubility.

The overall decrease in $\Delta S^{\ominus}_{surroundings}$ down the group is 368 $J\,K^{-1}\,mol^{-1}$ (+ 305 to −63).

The overall increase in the standard entropy change of the hydrated cations is 148 $J\,K^{-1}\,mol^{-1}$ (−138 to +10).

As the decrease in the standard entropy of the hydrated cations is much less than the increase in the entropy of the surroundings, the Group 2 metal sulfates become less soluble as you go down the group.

LEARNING TIP

At this level, it is assumed that ΔS_{system} does not change with temperature, since the entropies of both the reactants and the products change by similar amounts.

The magnitude of $\Delta S_{surroundings}$ will always decrease with increasing temperature. If $\Delta S_{surroundings}$ is negative, it will become less negative. If $\Delta S_{surroundings}$ is positive, it will become less positive.

Evidence from thermodynamics does not indicate the rate at which a reaction will occur. A reaction that is thermodynamically spontaneous at a given temperature may have a high activation energy. This could result in the reaction not taking place at all.

CHECKPOINT

1. Predict whether there is likely to be an increase, a decrease or no change in the entropy of the system in each of the following reactions. In each case, give your reasons.

(a) $CuSO_4.5H_2O(s) \rightarrow CuSO_4(s) + 5H_2O(l)$

(b) $HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$

(c) $SO_2(g) + \frac{1}{2}O_2(g) \rightarrow SO_3(g)$

(d) $Co(H_2O)_6^{2+}(aq) + EDTA^{2-}(aq) \rightarrow Co(EDTA)(aq) + 6H_2O(l)$

SKILLS CREATIVITY, PROBLEM-SOLVING

2. This question is about the following reaction:

$$H_2(g) + I_2(g) \rightarrow 2HI(g)$$

(a) State why you might predict that the entropy change in the system for this reaction is zero.

(b) The actual ΔS_{system} for this reaction is +22 $J\,K^{-1}\,mol^{-1}$. Explain why the total entropy change for this reaction is not zero.